CDA 106
Numerical Methods for Data Science

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QR Decomposition

$$A_{m \times k} = \underbrace{Q_{m \times k}}_{\text{orthogonal}} \underbrace{R_{k \times k}}_{\text{upper triangular}}$$

orthogonal $\quad QQ^T = I_m$

$\qquad\qquad\quad Q^TQ = I_k$

$$Q = \left[q_1 \,\middle|\, q_2 \,\middle|\, \cdots, q_n \right]$$

$$Q^T = \begin{bmatrix} q_1^T \\ \hline q_2^T \\ \hline \vdots \\ \hline q_n^T \end{bmatrix}$$

$$q_i = \begin{bmatrix} q_{1i} \\ q_{2i} \\ q_{3i} \\ q_{ni} \end{bmatrix} \quad \forall i$$

$$Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix}_{n \times 1} \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}_{1 \times n}$$

$$= \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ - & - & & - \\ - & & & q_n^T q_n \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = I_n$$

$Q \rightarrow$ orthogonal matrix

$$\langle q_i \, q_j \rangle = q_i^T q_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Example

$$Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & \sqrt{2/3} \\ 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}_{3\times 2} = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$$

$$Q^T Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

$$= I_2$$

Complete calculations & verify !

For a system $Ax = b$ — (1)

— $\text{rank}(A) = \text{rank}(A|b)$

then (1) has solution $\begin{cases} \text{unique} \\ \text{inf. many.} \end{cases}$

unique — $\text{rank}(A) =$ no of variables / no. of columns

inf. — $\text{rank}(A) <$ no of variable

Otherwise <u>no sol^n</u>

Recall

System $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$A \quad x \quad b$

Using QR decomposition, we found $x = 2/3$, $y = 2/3$ sol^n

$\boxed{\text{LSS}}$

In this case, $\text{rank}(A) = 2$

$[A|b] = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{array}\right]$

$\text{rank}(A|b) = 3$

$\text{rank}(A) \neq \text{rank}(A|b)$

$\text{Det}([A|b]) \neq 0$

Therefore no sol^n.

— We shall try and understand meaning of the <u>solution</u> of <u>$AX = b$</u> (via QR decomposition)

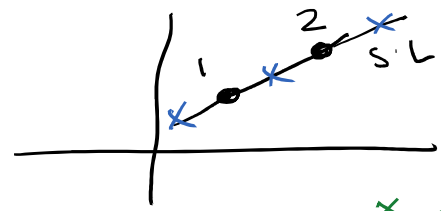
— Why do we need such solution?

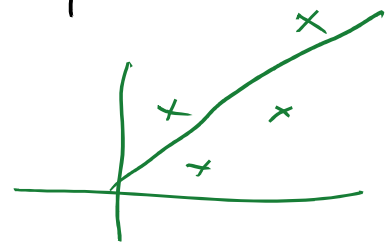
<u>Answer</u> Real life you will come across system of eqns where more eqn are given in lesser number of variables.

Let's consider the following example.

We all know that if we want to fit a straight line we need a minimum of two (2) points.

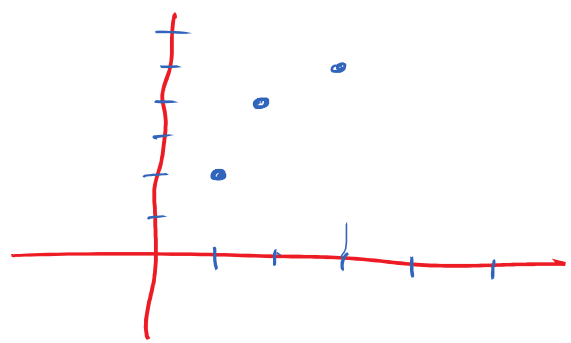
— However, what ~~will~~ if we need to fit a straight line ~~which~~ through more than 2 point





Let us consider following data points

$(1, 2)$; $(2, 4)$, $(3, 5)$



I wants to fit a line from these data points $y = \underbrace{m}_{\text{unknown}} x + \underbrace{c}_{\text{unknown}}$

(*) These kind of problem ~~arise~~ arise in curve fitting / Least square sol^n.

I should expect that the fitted line passes through all the data points, i.e.

all the points $(1,2)$, $(2,4)$ & $(3,5)$ satisfy

$$y = mx + c.$$

i.e.

$$\left.\begin{aligned} 2 &= m + c \quad &(1)\\ 4 &= 2m + c \quad &(2)\\ 5 &= 3m + c \quad &(3)\end{aligned}\right\} \text{ system of eqns}$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}}_{b}$$

Eventually, the fitting problem boils down to solⁿ of

$$Ax = b.$$

$\text{rank}(A) = 2$, $\text{rank}(A|b) = 3$

Since $\text{rank}(A) \neq \text{rank}(A/b)$ we do not have exact sol^n i.e. line passing through all the points.

However, we can find best sol^n $\hat{x}$ such that the solution $\hat{x}$ is closest to all the points.

[x cap or x hat $\hat{x}$]

Best Sol^n / Least Square Solution (LSS)

To find least square solution, we write $AX = b$ in form of

$$\boxed{A^T A \hat{X} = A^T b} \quad \text{——} \quad (*)$$

Instead of solving $AX = b$ we solve $A^T A \hat{X} = A^T b$ & $\hat{X}$ is known as LSS & $(*)$ is called normal eq^n.

A^TA is symmetric matrix

Because $(A^TA)^T = A^T(A^T)^T = A^TA$

In our case, LSS is given by

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}}_{b} \Rightarrow \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}}_{b}$$

$$\Rightarrow \underbrace{\begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}}_{(A^TA)} \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_{\hat{x}} = \begin{bmatrix} 25 \\ 11 \end{bmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \boxed{\begin{matrix} m = 3/2 \\ c = 2/3 \end{matrix}}$$

$\downarrow$

LSS

Rough

$$\begin{aligned} 14m + 6c &= 25 \\ 6m + 3c &= 11 \times 2 \\ \hline 2m &= 3 \end{aligned}$$

$$\boxed{y = \frac{3}{2}x + \frac{2}{3}}$$

Least square solution can be found via

$\underline{Ax = b}$

(1) Normal Method.

$$A^T A \hat{x} = A^T b$$

(2) QR decomposition $\quad Ax = b \Rightarrow QR\hat{x} = b$

$$\boxed{R\hat{x} = Q^T b}$$

$\underline{\underline{L.S.S.}}$

Total error

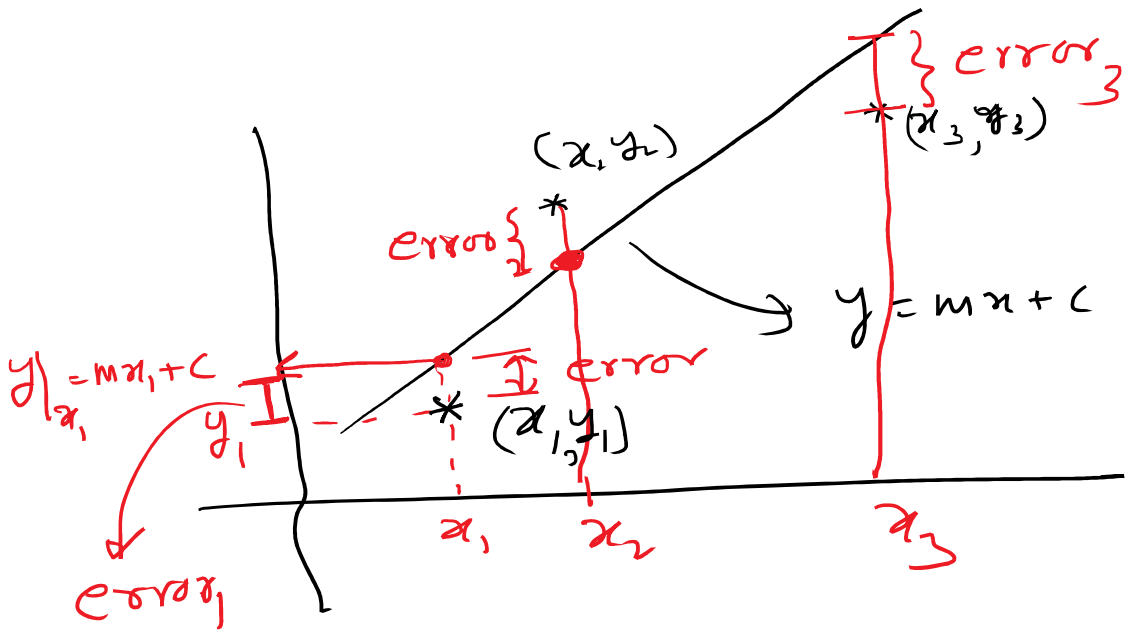
$= \text{error}_1 + \text{error}_2 + \text{error}_3$

$= (y|_{x_1} - y_1)^2$

$+ (y|_{x_2} - y_2)^2 + (y|_{x_3} - y_3)^2$

= sum of squares of all the errors

$$E = \sum_{i=1}^{N} \left(y_i - y|_{x_i}\right)^2$$



Best line $y = mx + c$

$E \longrightarrow \underline{\underline{\min}}$

Max / Min calculus

In general $\quad Ax = b \quad$ (sol^n)

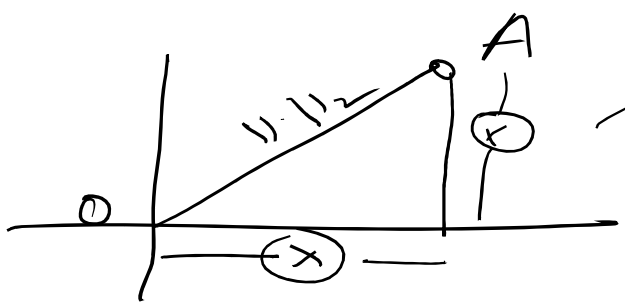
* we find $\hat{x}$ which gives sol^n

$$\text{error} = \underbrace{\|A\hat{x} - b\|_2}_{\text{LSS}} \longrightarrow \underline{\text{minimum}}$$

$\|\cdot\|_2 \longrightarrow$ Euclidean norm

$\|\cdot\|_1 \longrightarrow$ mod.

$\|\cdot\|_\infty \longrightarrow$ Max.



Let us consider now an example for <u>QR decomposition</u>

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}_{4\times 3}$$

$$\begin{array}{ccc} a_1 & a_2 & a_3 \end{array}$$

<u>Aim</u> To find $\underline{Q_{4\times 3}\, R_{3\times 3}}$

$$A = \left[a_1 \middle| a_2 \middle| a_3 \right]_{4\times 3} = \left[e_1 \middle| e_2 \middle| e_3 \right]_{4\times 3} \begin{bmatrix} a_1^T e_1 & a_2^T e_1 & a_3^T e_1 \\ 0 & a_2^T e_2 & a_3^T e_2 \\ 0 & 0 & a_3^T e_3 \end{bmatrix}_{3\times 3}$$

$$\langle a, b \rangle = a^T b$$

Using Gram Schmidt

$$u_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\|a_1\| = \sqrt{4} = 2$$

$$\boxed{e_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}}$$

$$\langle a_2, e_1 \rangle = a_2^T e_1 = \begin{bmatrix} 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$= 0$$

$$u_2 = a_2 - \langle a_2, e_1 \rangle e_1$$

$$u_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} - 0 \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\|u_2\| = \sqrt{4} = 2$$

$$\boxed{e_2 = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}}$$

$$u_3 = a_3 - \langle a_3, e_1 \rangle e_1 - \langle a_3, e_2 \rangle e_2$$

$$\langle a_3, e_1 \rangle = a_3^T e_1 = \begin{bmatrix} 0 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = 1$$

$$\langle a_3, e_2 \rangle = a_3^T e_2 = \begin{bmatrix} 0 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} = -2$$

$$u_3 = \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} + 2 \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

$$\|u_3\| = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4}} = 1$$

$$\boxed{e_3 = \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}}$$

$$\langle a_1, e_1 \rangle = a_1^T e_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = 2$$

$$\langle a_2, e_2 \rangle = a_2^T e_2 = \begin{bmatrix} 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} = 2$$

$$\langle a_3, e_3 \rangle = a_3^T e_3 = \begin{bmatrix} 0 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix} = 1$$

$$A = \underbrace{\begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \end{bmatrix}}_{\begin{matrix} e_1 & e_2 & e_3 \end{matrix}} \underbrace{\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}}_{R}$$

$$\underbrace{\qquad\qquad}_{Q}$$

Now, let's ~~solve~~ Find sol^n $A\textcircled{x} = b$ s.t.

$$\underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}}_{\textcircled{b}}$$

<u>QR decomposition</u>

LSS / Best sol^n

$\hat{x}$ s.t. $\underline{\underline{R\hat{x} = Q^T b}}$

For LSS

$$R\hat{x} = Q^T b$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$$

Backward Substitution

$$\boxed{z = 0}$$

$$2y - 2z = -1 \Rightarrow \boxed{y = -1/2}$$

$$2x = 4 \Rightarrow \boxed{x = 2}$$

$$\boxed{\hat{x} = \begin{bmatrix} 2 \\ -1/2 \\ 0 \end{bmatrix}}$$

LSS

We have ~~f~~ discussed the necessity and meaning of LSS and then found ways to evaluate it using Normal Eqn (Method) and QR decomposition method.

We also understood how the LSS is associated with error and it's minimization.

——— x ——— x ———